

# FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

## *Student Assessment Document*

### FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10: COMMERCIAL ARITHMETIC 1 (SUB-STRAND 2.8) FINAL EXPLANATION

#### Student Assessment Document

|              |       |
|--------------|-------|
| Student Name | _____ |
| Class        | _____ |
| Date         | _____ |

#### INSTRUCTIONS FOR STUDENTS

Amina is a Form 4 student from Kisumu. She saved KSh 4,500 from selling mandazi at school. She wants to use it to (1) contribute to the Environmental Club's end-of-term trip to Lake Victoria, (2) buy a second-hand smartphone at a Kondele electronics stall advertised at 20% off, and (3) send the remaining money to her aunt in Uganda through a mobile-money agent. Use what you learned in Lessons 1 to 8 to answer the questions below. Write your answers in the space under each question. Show every step of your working, write the formula you use, and include units (KSh or UGX). Explain your reasoning in full sentences where the question asks you to explain. Your final answer should show how commercial arithmetic helps Amina make smart, fair and informed money decisions.

#### Part 1: Why Amina needs a budget

The Environmental Club plans a one-day trip to Lake Victoria. Its total expenditure is KSh 12,250 and its income is KSh 6,000. (a) Calculate the difference between the club's income and expenditure, and state what this difference is called. (b) The cost list includes bus hire and entry fees. Say which is a fixed cost and which is a variable cost, and explain how you can tell. (c) Amina has not yet written down her own costs. Using the club's situation as evidence, explain why she should prepare a budget before she starts spending her KSh 4,500.

Write your answer here:

### Part 2: Discounts and which deal is better

Amina first gives KSh 800 to the Environmental Club, which leaves her with KSh 3,700. (a) The Kondele stall advertises the phone at a marked price of KSh 3,200 with 20% off. Calculate the discount and the sale price, and say whether Amina can afford it from her KSh 3,700. How much would she have left? (b) A Naivas supermarket advertises a similar phone at a marked price of KSh 3,440 with 25% off. Calculate its sale price. (c) Amina says: 'Naivas has the bigger percentage discount, so it must be the better deal.' Use your answers to decide whether she is correct, and explain how to compare discount offers fairly. (d) Explain why, in your calculation of the 20% discount, you were checking the advertisement with mathematics rather than just trusting the sign.

*Write your answer here:*

### Part 3: Commission and crossing the border

Amina has KSh 1,140 left to send to her aunt in Kampala. The mobile-money agent at Kondele charges a commission of 3% of the amount sent. (a) Calculate the commission the agent earns and the amount that is left to be converted. (b) The bureau's buying rate is 27.80 UGX per KES. Calculate how many Uganda Shillings her aunt receives, to the nearest shilling. (c) Amina saw a rate of 28.50 UGX per KES yesterday and expected to use that. Explain whether it is the right rate to use for her transaction and why. (d) Explain how the forex bureau's buying and selling rates can work like a hidden commission, and compare this with the agent's visible commission.

*Write your answer here:*

#### **Part 4: Profit, loss and fair prices**

Every price Amina meets in the market has a profit margin built into it. Suppose the Kondele stall paid KSh 2,000 for the phone and sold it to Amina at the discounted price you found in Part 2. (a) Calculate the stall's profit. (b) Calculate the percentage profit, and state which amount the percentage is calculated on. (c) Does the stall make a loss by offering the 20% discount? Explain what this tells Amina about whether the discount is fair to both her and the trader. (d) Amina's own mandazi business also has costs. Explain why she should use her total cost price, including overheads such as fuel, when she works out her percentage profit.

*Write your answer here:*

### **Part 5: Amina's complete financial plan and the driving question**

**Bring all your work together. (a) Write Amina's complete plan as a short budget starting from her KSh 4,500. Show the three allocations (the club contribution, the phone, the Uganda transfer), the commission, and what her aunt receives in UGX. (b) The same KSh 4,500 seems to produce different outcomes depending on discounts, commissions, exchange rates and profit or loss. Explain how each of these changes what Amina ends up with. (c) Answer the two key inquiry questions in your own words: Why do we need a budget? Why are discounts and commissions necessary in trade? (d) Answer the driving question: How can Amina, and every Kenyan, use commercial arithmetic to make smart, fair and informed money decisions in markets, clubs and across borders?**

Write your answer here:

| RUBRIC                        |                                                                                                                                                                                                                                   |                                                                                                                                                   |                                                                                                                                                      |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Criterion                     | Excellent (4)                                                                                                                                                                                                                     | Proficient (3)                                                                                                                                    | Developing (1–2)                                                                                                                                     |
| Accuracy of calculations      | All calculations (deficit, discount, sale price, commission, UGX conversion, profit and percentage profit) are correct, use the right formulas and the right base amounts, and carry through correctly from one step to the next. | Most calculations are correct and use the right formulas, with one or two minor arithmetic or rounding slips that do not change the overall plan. | Several calculations are incorrect or formulas are missing or misused, for example using the wrong base for a percentage or the wrong exchange rate. |
| Use of concepts and terms     | Uses terms such as fixed and variable cost, deficit, marked price, sale price, commission, buying rate, selling rate, spread, cost price and percentage profit correctly and precisely throughout.                                | Uses most terms correctly, with occasional imprecision.                                                                                           | Terms are missing, confused or used incorrectly.                                                                                                     |
| Reasoning and explanation     | Explains clearly why percentages cannot be compared alone, why the exchange rate depends on direction, why the spread is a hidden commission, and why percentage profit uses cost price, with evidence from the calculations.     | Gives mostly correct explanations, but some lack evidence or full reasoning.                                                                      | Explanations are missing, vague or incorrect, or are not supported by the calculations.                                                              |
| Integration of the whole plan | Presents a complete, consistent plan in which each step feeds into the next                                                                                                                                                       | Presents a mostly complete plan with the main steps linked, but minor gaps in the                                                                 | The plan is incomplete or the steps are not linked, and little connection is made                                                                    |

|                                 |                                                                                                                                   |                                                                     |                                                                                  |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------|
|                                 | (budget, discount, commission, conversion) and accounts for all KSh 4,500, and links it clearly to the driving question.          | connections or the link to the driving question.                    | to the driving question.                                                         |
| Kenyan context and presentation | Keeps Amina's Kenyan context central, shows working clearly with units and steps, and writes in an organised, easy-to-follow way. | Working is mostly clear with units, and the context is mostly used. | Working is hard to follow, units are often missing, and the context is not used. |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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